Congruence and Sylvester's Law of Inertia

An Elementary Exposition with Applications In Mathematical Statistics

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Abstract

In many modern undergraduate linear algebra curricula, the concept of *inertia* receives relatively little attention, if it is discussed at all. Consequently, students may become familiar with the diagonalization of symmetric matrices and the notion of matrix similarity without fully appreciating the distinction between similarity and congruence. This distinction, although subtle at first glance, is fundamental to the theory of quadratic forms.

In this note, we aim to clarify these ideas and develop Sylvester's Law of Inertia from first principles. Beginning with quadratic forms and changes of basis, we introduce matrix congruence, define the inertia of a real symmetric matrix, and establish the remarkable fact that the numbers of positive, negative, and zero eigenvalues remain unchanged under congruence.

Particular attention is devoted to a detailed, self-contained proof of the theorem, emphasizing the underlying linear algebra rather than merely stating the result.

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1 Introduction

One of the fundamental results of linear algebra is the spectral theorem, which guarantees that every real symmetric matrix can be diagonalized by an orthogonal change of basis. While diagonalization and matrix similarity are familiar topics in undergraduate linear algebra, the closely related notion of *congruence* often receives less attention.

Similarity and congruence arise in different mathematical settings. The transformation $S^{-1}AS$ describes a change of basis for a linear operator, whereas S^TAS arises naturally from a change of coordinates in a quadratic form. Indeed, if a real symmetric matrix A defines

$$q(x) = x^TAx,$$

then the substitution $x = Sy$, with S invertible, gives

$$q(x) = y^T(S^TAS)y.$$

Thus, A and S^TAS represent the same quadratic form in different coordinate systems and are said to be *congruent*.

Unlike similarity, congruence does not generally preserve individual eigenvalues. Remarkably, however, it preserves the numbers of positive, negative, and zero eigenvalues. These three numbers constitute the *inertia* of a real symmetric matrix, and their invariance under congruence is known as *Sylvester's Law of Inertia*.

The purpose of this note is to develop this theorem from first principles. We begin with a brief review of quadratic forms and changes of basis, distinguish congruence from similarity, and introduce the notions of inertia, rank, index, and signature. We then present a detailed proof of Sylvester's Law of Inertia using elementary tools from linear algebra, followed by its canonical form and applications in mathematical statistics and theoretical physics. We conclude with a brief historical account of James Joseph Sylvester and the development of the theorem.

2 Prerequisites

This note is intended for students and researchers who have completed a first course in linear algebra. Readers are expected to be well versed in eigenvalues and eigenvectors, eigendecomposition, the spectral theorem for real symmetric matrices, and the theory of quadratic forms.

Familiarity with changes of basis, orthonormal bases, matrix rank, null spaces, and the rank-nullity theorem is also assumed. In particular, the reader should understand why real symmetric matrices admit orthonormal eigenbases and how a quadratic form can be represented by a symmetric matrix.

Although these concepts form the mathematical foundation of our discussion, the essential ideas will be briefly revisited whenever necessary. No knowledge of Sylvester's Law of Inertia or the theory of matrix congruence is assumed.

3 Quadratic Forms and Changes of Basis

Before introducing matrix congruence, we briefly recall the notion of a quadratic form and examine how its matrix representation changes under a change of basis.

3.1 Quadratic Forms

A quadratic form on $\mathbb{R}^n$ is a scalar-valued function of the form

$$q(x) = x^T Ax,$$

where $A \in \mathbb{R}^{n\times n}$ is a real symmetric matrix and $x \in \mathbb{R}^n$ is a column vector.

For example, consider

$$q(x_1, x_2) = 2x_1^2 + 2x_1x_2 + 3x_2^2.$$

This quadratic form can be represented as

$$q(x) = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Notice that the coefficient of the mixed term x_1x_2 is distributed equally between the corresponding off-diagonal entries of the symmetric matrix.

More generally, every quadratic form on $\mathbb{R}^n$ admits a unique symmetric matrix representation relative to a fixed basis.

3.2 Changing the Basis

Suppose that the quadratic form

$$q(x) = x^T Ax$$

is initially expressed in the standard basis of $\mathbb{R}^n$.

Let

$$\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$$

be another basis, and define the change-of-basis matrix

$$S = \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix}.$$

Since the columns of S form a basis, S is invertible.

If y denotes the coordinate vector of x relative to the new basis $\mathcal{B}$, then

$$x = Sy.$$

Substituting into the quadratic form gives

$$\begin{aligned} q(x) &= x^T A x \\ &= (Sy)^T A (Sy) \\ &= y^T S^T A S y. \end{aligned}$$

Therefore, if we define

$$\boxed{B = S^T A S},$$

and the quadratic form in the new coordinates becomes

$$q(Sy) = y^T B y.$$

Thus, A and B represent the same quadratic form relative to two different bases.

This motivates the following definition.

Definition 3.1 (Matrix Congruence)**.** Two real symmetric matrices $A, B \in \mathbb{R}^{n\times n}$ are said to be *congruent* if there exists an invertible matrix S such that

$$\boxed{B = S^T A S}.$$

Remark 3.2. A change of basis does not alter the quadratic form itself; it changes only its matrix representation. Moreover, the change-of-basis matrix need not be orthogonal. Any invertible matrix is permitted.

3.3 An Illustrative Example

Consider the quadratic form

$$q(x_1, x_2) = 2x_1^2 + 2x_1x_2 + 3x_2^2,$$

whose symmetric matrix representation is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Suppose we introduce a new basis

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The corresponding change-of-basis matrix is

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The matrix of the quadratic form in the new basis is

$$\begin{aligned} B &= S^T A S \\ &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 2 & 3 \\ 3 & 7 \end{pmatrix}}. \end{aligned}$$

To verify this directly, observe that $x = Sy$ implies

$$x_1 = y_1 + y_2, \qquad x_2 = y_2.$$

Substituting into the original quadratic form,

$$\begin{aligned} q(x) &= 2(y_1 + y_2)^2 + 2(y_1 + y_2)y_2 + 3y_2^2 \\ &= 2y_1^2 + 6y_1 y_2 + 7y_2^2 \\ &= \begin{pmatrix} y_1 & y_2 \end{pmatrix} \boxed{\begin{pmatrix} 2 & 3 \\ 3 & 7 \end{pmatrix}} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}. \end{aligned}$$

We have therefore obtained two different symmetric matrices representing exactly the same quadratic form in different coordinate systems.

The important question is now the following: although congruence preserves the quadratic form, does it also preserve its eigenvalues?

As we shall see in the next section, the answer is generally no. This distinction separates congruence from matrix similarity and ultimately leads to Sylvester's Law of Inertia.

4 Similarity Versus Congruence

Although similarity and congruence are both associated with changes of basis, they describe fundamentally different mathematical relationships. It is important to distinguish them before proceeding to Sylvester's Law of Inertia.

4.1 Similarity and Linear Operators

Two matrices $A, B \in \mathbb{R}^{n \times n}$ are said to be *similar* if there exists an invertible matrix S such that

$$B = S^{-1} A S.$$

Similar matrices represent the same linear operator relative to different bases. More importantly, they have the same characteristic polynomial:

$$\begin{aligned} \det(\lambda I - B) &= \det(\lambda I - S^{-1} A S) \\ &= \det\big(S^{-1}(\lambda I - A) S\big) \\ &= \det(\lambda I - A). \end{aligned}$$

Consequently, similar matrices possess exactly the same set of eigenvalues, including their algebraic multiplicities. In particular, similarity preserves the trace, determinant, and rank.

4.2 Congruence and Quadratic Forms

Recall from the preceding section that two real symmetric matrices A and B are congruent if

$$B = S^T A S$$

for some invertible matrix S.

Unlike similarity, congruence concerns the matrix representation of a quadratic form rather than that of a linear operator, and it preserves the rank of a matrix, though, it does not, in general, preserve its eigenvalues. The following example illustrates this distinction.

Example 4.1 (Congruence Does Not Preserve Eigenvalues)**.** Consider the two real symmetric matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 0 \\ 0 & -18 \end{pmatrix}.$$

We first observe that the eigenvalues of A are 1 and -2, whereas those of B are 4 and -18. Since similar matrices must have identical eigenvalues, A and B cannot be similar.

However, are these matrices congruent?

To answer this question, recall that congruence requires the existence of an invertible matrix S such that

$$B = S^T A S.$$

Let us look for a diagonal matrix of the form

$$S = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \qquad ab \neq 0.$$

Then

$$\begin{aligned} S^T A S &= \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \\ &= \begin{pmatrix} a^2 & 0 \\ 0 & -2b^2 \end{pmatrix}. \end{aligned}$$

For this matrix to equal B, we require

$$a^2 = 4, \qquad -2b^2 = -18.$$

One possible choice is $a = 2$ and $b = 3$. Therefore, taking

$$S = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

we obtain

$$\boxed{S^T A S = B}.$$

Furthermore,

$$\det(S) = 6 \neq 0,$$

so S is invertible. Hence, by the definition of matrix congruence, A and B are congruent.

We have thus exhibited two matrices that are congruent but not similar.

Interestingly, although their eigenvalues differ, both matrices possess exactly one positive and one negative eigenvalue.

This observation anticipates Sylvester's Law of Inertia: congruence does not preserve the individual eigenvalues, but it does preserve the numbers of positive, negative, and zero eigenvalues.

Remark 4.2. When S is orthogonal, we have $S^{-1} = S^T$. In this special case, congruence and similarity take the same form:

$$S^T A S = S^{-1} A S.$$

Consequently, orthogonal congruence preserves the eigenvalues. For general invertible matrices, however, this need not be true.

4.3 Why Rank Alone Is Not Enough

Since multiplication by an invertible matrix does not change rank, congruent matrices necessarily have the same rank:

$$\begin{aligned} \operatorname{rank}(S^T A S) &= \operatorname{rank}(AS) \\ &= \operatorname{rank}(A). \end{aligned}$$

But is equality of ranks sufficient to guarantee congruence?

Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Both matrices have rank one. Nevertheless, they are not congruent.

To see why, observe that

$$x^T A x = x_1^2 \geq 0$$

for every $x \in \mathbb{R}^2$, whereas

$$x^T B x = -x_1^2 \leq 0.$$

Indeed, the quadratic form associated with B takes strictly negative values, while the one associated with A never does.

If $B = S^T A S$ for some invertible S, then

$$x^T B x = (Sx)^T A (Sx) \geq 0$$

for every x, contradicting the preceding observation.

Therefore, equal rank is necessary but not sufficient for congruence.

The obstruction in this example arises from the signs of the quadratic forms, or equivalently, from the signs of their eigenvalues.

This motivates the central question of our note: *What information about the eigenvalues of a real symmetric matrix remains invariant under congruence?*

The answer leads naturally to the concept of *inertia* and, ultimately, to Sylvester's Law of Inertia.

5 Inertia, Rank, Index, and Signature

In the preceding section, we observed that congruent matrices need not have identical eigenvalues. Nevertheless, the signs of their eigenvalues appear to carry important information about the underlying quadratic form.

This observation motivates the concept of *inertia*, which plays a central role in the classification of real symmetric matrices.

5.1 Definition of Inertia

Let $A \in \mathbb{R}^{n\times n}$ be a real symmetric matrix. By the spectral theorem, all eigenvalues of A are real, and there exists an orthogonal matrix Q such that

$$Q^T AQ = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

We may arrange these eigenvalues so that the positive eigenvalues appear first, followed by the negative eigenvalues, and finally the zero eigenvalues.

Definition 5.1 (Inertia of a Symmetric Matrix)**.** Let n_+, n_-, and n_0 denote the numbers of positive, negative, and zero eigenvalues of A, respectively, counted with their multiplicities.

The *inertia* of A is the ordered triple

$$\boxed{\operatorname{In}(A) = (n_+, n_-, n_0)}.$$

Since every eigenvalue belongs to exactly one of these three categories, we necessarily have

$$\boxed{n_+ + n_- + n_0 = n}.$$

The inertia therefore describes the distribution of eigenvalues according to their signs, rather than their actual numerical values.

5.2 Rank, Index, and Signature

Several important quantities can be expressed directly in terms of inertia.

Definition 5.2 (Rank, Index, and Signature)**.** For a real symmetric matrix A, we define:

1. The *rank* r is the number of nonzero eigenvalues:
$$r = n_+ + n_-.$$
2. The *index* p is the number of positive eigenvalues:
$$p = n_+.$$
3. The *signature* s is the number of positive eigenvalues minus the number of negative eigenvalues:
$$s = n_+ - n_-.$$

Here, the term *index* refers specifically to the positive index of inertia. Some authors use different terminology, but we shall follow this convention throughout the note.

Since
$$r = n_+ + n_-$$
and $p = n_+$, the number of negative eigenvalues must be
$$n_- = r - p.$$

Consequently, the signature becomes
$$\begin{aligned} s &= n_+ - n_- \\ &= p - (r - p) \\ &= 2p - r. \end{aligned}$$

Thus,
$$\boxed{s = 2p - r}.$$

Equivalently, if the rank and signature are known, we can recover the numbers of positive and negative eigenvalues:
$$\boxed{n_+ = \frac{r+s}{2}, \qquad n_- = \frac{r-s}{2}}.$$

Furthermore, by the rank-nullity theorem,
$$n_0 = n - r.$$

We therefore obtain
$$\boxed{\mathrm{In}(A) = (p, r-p, n-r)}.$$

This shows that, for a symmetric matrix of fixed size, its rank and index completely determine its inertia. The signature is not an independent quantity.

5.3 An Illustrative Example

Consider the real symmetric matrix

$$A = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Since A is already diagonal, its eigenvalues are immediately obtained:

$$\lambda_1 = 5, \qquad \lambda_2 = 2, \qquad \lambda_3 = -3, \qquad \lambda_4 = 0.$$

There are two positive eigenvalues, one negative eigenvalue, and one zero eigenvalue. Hence,

$$\boxed{\mathrm{In}(A) = (2, 1, 1)}.$$

The corresponding rank, index, and signature are

$$\begin{aligned} r &= 2 + 1 = 3, \\ p &= 2, \\ s &= 2 - 1 = 1. \end{aligned}$$

As expected, the identity $s = 2p - r$ gives

$$s = 2(2) - 3 = 1.$$

5.4 Geometric Interpretation

The inertia of a symmetric matrix also describes important geometric properties of its associated quadratic form.

To see this, suppose that

$$A = QDQ^T,$$

where Q is orthogonal and

$$D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Introducing the orthogonal coordinates $y = Q^T x$, we obtain

$$\begin{aligned} q(x) &= x^T A x \\ &= x^T Q D Q^T x \\ &= y^T D y \\ &= \sum_{i=1}^{n} \lambda_i y_i^2. \end{aligned}$$

Thus, in an orthonormal eigenbasis, every quadratic form becomes a sum of squared coordinates weighted by its eigenvalues.

Positive eigenvalues contribute positively to the quadratic form, negative eigenvalues contribute negatively, and zero eigenvalues make no contribution.

For instance, the matrix in our previous example defines

$$q(x) = 5x_1^2 + 2x_2^2 - 3x_3^2.$$

The form is strictly positive for every nonzero vector in the subspace

$$\text{span}\{e_1, e_2\},$$

and strictly negative for every nonzero vector in

$$\text{span}\{e_3\}.$$

Moreover, the vector e_4 belongs to the null space of A, so

$$Ae_4 = 0.$$

These subspaces have dimensions two, one, and one, respectively, precisely matching the inertia $(2, 1, 1)$.

Remark 5.3. A vector satisfying $q(x) = 0$ need not belong to the null space of A. For example, if $A = \text{diag}(1, -1)$, then $q(1, 1) = 0$, although $A(1, 1)^T \neq 0$. The zero component of inertia counts zero eigenvalues, or equivalently, the dimension of $\ker(A)$.

We are now ready to formulate the central result of this note: Sylvester's Law of Inertia. Recall that the inertia of a real symmetric matrix is the ordered triple (n_+, n_-, n_0), recording the numbers of positive, negative, and zero eigenvalues, respectively. Sylvester's Law of Inertia asserts that these three numbers remain unchanged under congruence, even though the eigenvalues themselves may change.

More remarkably, the converse is also true: two real symmetric matrices of the same size are congruent if they have identical inertia. Thus, inertia provides a complete classification of real symmetric matrices under congruence. We shall prove both directions of this remarkable result in the next section.

6 Sylvester's Law of Inertia: Statement and Proof

As discussed in the preceding sections, congruence does not generally preserve the individual eigenvalues of a symmetric matrix. Nevertheless, it preserves the numbers of positive, negative, and zero eigenvalues. Sylvester's Law of Inertia states that these numbers completely characterize the congruence classes of real symmetric matrices.

6.1 Statement of the Theorem

Theorem 6.1 (Sylvester's Law of Inertia)**.** *Let $B, C \in \mathbb{R}^{n\times n}$ be real symmetric matrices. Then B and C are congruent if and only if they have the same inertia.*

Equivalently,

$$\boxed{B \text{ and } C \text{ are congruent} \iff \operatorname{In}(B) = \operatorname{In}(C).}$$

That is, there exists an invertible matrix S satisfying

$$C = S^T B S$$

if and only if B and C have the same numbers of positive, negative, and zero eigenvalues, counting multiplicities.

We will prove the theorem in two directions.

First, assuming that B and C are congruent, we will prove that their inertia must agree. This is the more challenging direction, and our proof will rely on the spectral theorem, orthonormal eigenbases, and a carefully constructed linear transformation.

We will then establish the converse by showing that two symmetric matrices with the same inertia can be transformed into the same diagonal matrix through congruence.

6.2 Congruence Implies Equal Inertia

Proof of the Forward Implication. **Step 1. Establish equality of ranks.**

Suppose that B and C are congruent. By definition, there exists an invertible matrix S such that

$$\boxed{C = S^T B S}.$$

Since S is invertible, so is S^T. Multiplication by an invertible matrix does not change rank. Therefore,

$$\begin{aligned}\operatorname{rank}(C) &= \operatorname{rank}(S^T B S)\\ &= \operatorname{rank}(BS)\\ &= \operatorname{rank}(B).\end{aligned}$$

Let their common rank be r. Thus,

$$\boxed{\operatorname{rank}(B) = \operatorname{rank}(C) = r.}$$

Because both matrices are $n \times n$, they also have the same nullity:

$$\dim\ker(B) = \dim\ker(C) = n - r.$$

Consequently, B and C have the same number of zero eigenvalues.

Our remaining task is to prove that they have the same numbers of positive and negative eigenvalues.

Step 2. Assume that the indices differ.

Let p denote the number of positive eigenvalues of B, and let q denote the number of positive eigenvalues of C.

Suppose, for contradiction, that

$$p \neq q.$$

Since the roles of B and C can be interchanged, we may assume without loss of generality that

$$\boxed{p < q}.$$

Indeed, congruence is symmetric: if $C = S^T BS$, then

$$B = S^{-T} C S^{-1}.$$

Since both matrices have rank r, the number of negative eigenvalues of B is $r - p$, whereas that of C is $r - q$.

Our assumption $p < q$ therefore implies

$$r - p > r - q.$$

Thus, under our assumption, B has fewer positive eigenvalues but more negative eigenvalues than C.

We shall show that this is impossible if the matrices are congruent.

Step 3. Choose orthonormal eigenbases.

Since B is real symmetric, the spectral theorem guarantees the existence of an orthonormal eigenbasis

$$\{v_1, v_2, \ldots, v_n\}$$

such that

$$Bv_i = a_i v_i,$$

where a_i is the eigenvalue corresponding to v_i.

We arrange the eigenvectors according to the signs of their eigenvalues:

$$\begin{cases} a_i > 0, & 1 \leq i \leq p, \\ a_i < 0, & p+1 \leq i \leq r, \\ a_i = 0, & r+1 \leq i \leq n. \end{cases}$$

Likewise, since C is real symmetric, there exists an orthonormal eigenbasis

$$\{w_1, w_2, \ldots, w_n\}$$

such that

$$Cw_i = e_i w_i,$$

where

$$\begin{cases} e_i > 0, & 1 \le i \le q, \\ e_i < 0, & q+1 \le i \le r, \\ e_i = 0, & r+1 \le i \le n. \end{cases}$$

In particular,

$$\ker(B) = \operatorname{span}\{v_{r+1}, \ldots, v_n\},$$

and

$$\ker(C) = \operatorname{span}\{w_{r+1}, \ldots, w_n\}.$$

These relations follow because the eigenvectors corresponding to zero eigenvalues form an orthonormal basis for the null space of a symmetric matrix.

Before proceeding, recall an important consequence of congruence.

Since

$$C = S^T B S,$$

for any $x \in \mathbb{R}^n$, define

$$y = S^{-1} x.$$

Then $x = Sy$, and hence

$$\begin{aligned} y^T C y &= y^T S^T B S y \\ &= (Sy)^T B (Sy) \\ &= x^T B x. \end{aligned}$$

Therefore,

$$\boxed{x^T B x = y^T C y, \qquad x = Sy.}$$

This equality expresses the fact that congruent matrices represent the same quadratic form in different coordinates.

Notice that we must compare $x^T B x$ with $y^T C y$, where $x = Sy$, rather than automatically using identical coordinate vectors for both forms.

Step 4. Construct a linear transformation.

We now introduce the central construction of the proof.

Define the linear transformation

$$L : \mathbb{R}^n \longrightarrow \mathbb{R}^{p+(r-q)}$$

by

$$\boxed{L(x) = \begin{pmatrix} v_1^T Bx \\ \vdots \\ v_p^T Bx \\ w_{q+1}^T CS^{-1}x \\ \vdots \\ w_r^T CS^{-1}x \end{pmatrix}.}$$

Let us examine this construction carefully.

The first p components correspond to the eigenvectors associated with the positive eigenvalues of B.

The remaining $r-q$ components correspond to the eigenvectors associated with the negative eigenvalues of C.

Every component is linear in x. Therefore, L is a linear transformation.

Our objective is to find a vector x_0 satisfying

$$L(x_0) = 0,$$

while ensuring that

$$Bx_0 \neq 0.$$

Such a vector will allow us to eliminate the positive-eigenvalue contributions of B and the negative-eigenvalue contributions of C.

The existence of this vector is guaranteed by the rank-nullity theorem.

Step 5. Apply the rank-nullity theorem.

Since

$$L : \mathbb{R}^n \longrightarrow \mathbb{R}^{p+(r-q)},$$

we necessarily have

$$\operatorname{rank}(L) \leq p + (r - q).$$

By the rank-nullity theorem,

$$\dim \ker(L) + \operatorname{rank}(L) = n.$$

Consequently,

$$\begin{aligned} \dim \ker(L) &= n - \operatorname{rank}(L) \\ &\geq n - \big(p + r - q\big) \\ &= n - r + (q - p). \end{aligned}$$

But our assumption was $p < q$. Therefore,

$$q - p > 0.$$

Hence,

$$\boxed{\dim \ker(L) > n - r}.$$

Recall, however, that

$$\dim \ker(B) = n - r.$$

Thus,

$$\boxed{\dim \ker(L) > \dim \ker(B).}$$

This inequality is the key observation.

A vector space of dimension greater than $n - r$ cannot be contained entirely in a vector space of dimension $n - r$.

Therefore, $\ker(L)$ cannot be contained in $\ker(B)$.

Consequently, there exists a vector $x_0 \in \mathbb{R}^n$ such that

$$\boxed{x_0 \in \ker(L), \qquad x_0 \notin \ker(B).}$$

Equivalently,

$$\boxed{L(x_0) = 0, \qquad Bx_0 \neq 0.}$$

Now define

$$\boxed{y_0 = S^{-1}x_0}.$$

Since $C = S^T BS$, we obtain

$$\begin{aligned} Cy_0 &= CS^{-1}x_0 \\ &= S^T BSS^{-1}x_0 \\ &= S^T Bx_0. \end{aligned}$$

Because S^T is invertible and $Bx_0 \neq 0$, we conclude that

$$\boxed{Cy_0 \neq 0}.$$

We have therefore constructed two corresponding coordinate vectors, x_0 and $y_0 = S^{-1}x_0$, satisfying

$$L(x_0) = 0, \qquad Bx_0 \neq 0, \qquad Cy_0 \neq 0.$$

We shall now study the quadratic form in these two coordinate systems.

Step 6. Express x_0 in the eigenbasis of B.

Since $\{v_1, \ldots, v_n\}$ is an orthonormal basis for $\mathbb{R}^n$, we may write

$$x_0 = \sum_{i=1}^{n} c_i v_i.$$

Recall that $L(x_0) = 0$.

By the definition of L, its first p components must vanish:

$$v_k^T B x_0 = 0, \qquad 1 \le k \le p.$$

Substituting the expansion of x_0, we obtain

$$\begin{aligned} 0 &= v_k^T B x_0 \\ &= v_k^T B \left(\sum_{i=1}^{n} c_i v_i \right) \\ &= \sum_{i=1}^{n} c_i v_k^T B v_i. \end{aligned}$$

Since $Bv_i = a_i v_i$,

$$0 = \sum_{i=1}^{n} c_i a_i v_k^T v_i.$$

But the eigenvectors form an orthonormal basis, so

$$v_k^T v_i = \delta_{ki},$$

where δ_{ki} denotes the Kronecker delta.

Therefore, all terms vanish except the term corresponding to $i = k$. We obtain

$$\boxed{c_k a_k = 0}.$$

Since $a_k > 0$ for $1 \le k \le p$, we must have

$$\boxed{c_k = 0, \qquad 1 \le k \le p}.$$

Thus,

$$\boxed{x_0 = \sum_{i=p+1}^{n} c_i v_i.}$$

This is precisely what we intended: the vector x_0 has no components in the positive-eigenvalue directions of B.

Furthermore, because $Bx_0 \neq 0$, at least one coefficient c_i with $p + 1 \le i \le r$ must be nonzero.

Indeed,

$$Bx_0 = \sum_{i=p+1}^{r} c_i a_i v_i,$$

and this expression cannot vanish unless all coefficients in the indicated range are zero.

Step 7. Show that $x_0^T B x_0 < 0$.

Using the expansion obtained above, we compute

$$\begin{aligned} x_0^T B x_0 &= \left(\sum_{i=p+1}^{n} c_i v_i \right)^T B \left(\sum_{j=p+1}^{n} c_j v_j \right) \\ &= \sum_{i=p+1}^{n} \sum_{j=p+1}^{n} c_i c_j v_i^T B v_j. \end{aligned}$$

Since $Bv_j = a_j v_j$,

$$x_0^T B x_0 = \sum_{i=p+1}^{n} \sum_{j=p+1}^{n} c_i c_j a_j v_i^T v_j.$$

By orthonormality, $v_i^T v_j = \delta_{ij}$. Therefore,

$$x_0^T B x_0 = \sum_{i=p+1}^{n} a_i c_i^2.$$

Since $a_i = 0$ for $i > r$, this reduces to

$$\boxed{x_0^T B x_0 = \sum_{i=p+1}^{r} a_i c_i^2.}$$

Now observe that

$$a_i < 0, \qquad p+1 \le i \le r.$$

Also, at least one coefficient c_i in this range is nonzero, because $Bx_0 \neq 0$.

Consequently, every summand is nonpositive, and at least one summand is strictly negative.

Hence,

$$\boxed{x_0^T B x_0 < 0}. \tag{1}$$

Thus, the quadratic form represented by B takes a strictly negative value at x_0.

Step 8. Express y_0 in the eigenbasis of C.

Recall that

$$y_0 = S^{-1} x_0.$$

Since $\{w_1, \ldots, w_n\}$ is an orthonormal eigenbasis for C, we may write

$$y_0 = \sum_{i=1}^{n} d_i w_i.$$

Again, since $L(x_0) = 0$, the last $r - q$ components of $L(x_0)$ vanish:

$$w_k^T C S^{-1} x_0 = 0, \qquad q+1 \le k \le r.$$

Since $y_0 = S^{-1} x_0$, this becomes

$$w_k^T C y_0 = 0.$$

Substituting the expansion of y_0, we obtain

$$\begin{aligned} 0 &= w_k^T C y_0 \\ &= w_k^T C \left(\sum_{i=1}^{n} d_i w_i \right) \\ &= \sum_{i=1}^{n} d_i w_k^T C w_i. \end{aligned}$$

Since $C w_i = e_i w_i$,

$$0 = \sum_{i=1}^{n} d_i e_i w_k^T w_i.$$

By orthonormality,

$$w_k^T w_i = \delta_{ki}.$$

Therefore,

$$\boxed{d_k e_k = 0}.$$

But

$$e_k < 0, \qquad q+1 \le k \le r.$$

Hence,

$$\boxed{d_k = 0, \qquad q+1 \le k \le r.}$$

Consequently,

$$\boxed{y_0 = \sum_{i=1}^{q} d_i w_i + \sum_{i=r+1}^{n} d_i w_i.}$$

Thus, y_0 has no components in the negative-eigenvalue directions of C.

Furthermore, since $C y_0 \neq 0$, at least one coefficient d_i with $1 \le i \le q$ must be nonzero.

Step 9. Show that $y_0^T C y_0 > 0$.

Using the orthonormal expansion, we have

$$\begin{aligned} y_0^T C y_0 &= \left(\sum_{i=1}^{n} d_i w_i\right)^T C \left(\sum_{j=1}^{n} d_j w_j\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} d_i d_j w_i^T C w_j \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} d_i d_j e_j w_i^T w_j \\ &= \sum_{i=1}^{n} e_i d_i^2. \end{aligned}$$

But we already established that

$$d_i = 0, \qquad q+1 \le i \le r,$$

and

$$e_i = 0, \qquad r+1 \le i \le n.$$

Hence,

$$\boxed{y_0^T C y_0 = \sum_{i=1}^{q} e_i d_i^2.}$$

For every index in this sum,

$$e_i > 0.$$

Moreover, at least one coefficient d_i with $1 \le i \le q$ is nonzero, because $Cy_0 \neq 0$.

Therefore, every summand is nonnegative, and at least one is strictly positive.

We conclude that

$$\boxed{y_0^T C y_0 > 0}. \tag{2}$$

Thus, the quadratic form represented by C takes a strictly positive value at y_0.

Step 10. Obtain the contradiction.

Recall that

$$C = S^T B S$$

and

$$x_0 = S y_0.$$

Consequently,

$$\begin{aligned} y_0^T C y_0 &= y_0^T S^T B S y_0 \\ &= (S y_0)^T B (S y_0) \\ &= x_0^T B x_0. \end{aligned}$$

Therefore,

$$\boxed{y_0^T C y_0 = x_0^T B x_0}.$$

However, equations (1) and (2) established that

$$x_0^T B x_0 < 0$$

while

$$y_0^T C y_0 > 0.$$

Combining these results, we obtain

$$0 < y_0^T C y_0 = x_0^T B x_0 < 0,$$

which is impossible.

We have therefore reached a contradiction.

Hence, our assumption that the indices of B and C differ must be false. It follows that

$$\boxed{p = q}.$$

We already established that the matrices have the same rank r. Consequently, they have the same number of negative eigenvalues:

$$r - p = r - q.$$

They also have the same number of zero eigenvalues:

$$n - r.$$

Thus, the numbers of positive, negative, and zero eigenvalues are identical for both matrices.

Therefore,

$$\boxed{\text{In}(B) = \text{In}(C).}$$

This completes the proof that congruence implies equality of inertia.

□

6.3 Equal Inertia Implies Congruence

We now establish the converse of the theorem.

Suppose that two real symmetric matrices have the same inertia. We must show that there exists an invertible matrix relating them through congruence.

The central idea is that every real symmetric matrix can be transformed, by congruence, into a diagonal matrix whose entries belong to $\{1, -1, 0\}$.

Proof of the Converse Implication. **Step 1. Assume equality of inertia.**

Let $B, C \in \mathbb{R}^{n\times n}$ be real symmetric matrices satisfying

$$\mathrm{In}(B) = \mathrm{In}(C).$$

Suppose their common inertia is

$$(p, r-p, n-r).$$

Thus, each matrix has p positive eigenvalues, $r-p$ negative eigenvalues, and $n-r$ zero eigenvalues.

Our objective is to construct an invertible matrix S such that

$$C = S^T B S.$$

Step 2. Diagonalize the matrix B.

By the spectral theorem, there exists an orthogonal matrix U such that

$$U^T B U = \mathrm{diag}(a_1, \ldots, a_n),$$

where the eigenvalues are ordered according to their signs:

$$\begin{cases} a_i > 0, & 1 \le i \le p, \\ a_i < 0, & p+1 \le i \le r, \\ a_i = 0, & r+1 \le i \le n. \end{cases}$$

We now introduce an invertible diagonal matrix that will normalize the nonzero eigenvalues of B.

Define

$$D_B = \mathrm{diag}\left(\frac{1}{\sqrt{|a_1|}}, \ldots, \frac{1}{\sqrt{|a_r|}}, 1, \ldots, 1\right).$$

The entries corresponding to zero eigenvalues are chosen to be 1, rather than attempting to divide by zero.

Since every diagonal entry of D_B is nonzero, D_B is invertible.

Now observe that

$$\begin{aligned} D_B^T U^T B U D_B &= D_B^T \,\mathrm{diag}(a_1, \ldots, a_n) D_B \\ &= \mathrm{diag}\left(\frac{a_1}{|a_1|}, \ldots, \frac{a_r}{|a_r|}, 0, \ldots, 0\right). \end{aligned}$$

But

$$\frac{a_i}{|a_i|} = \begin{cases} 1, & a_i > 0, \\ -1, & a_i < 0. \end{cases}$$

Therefore,

$$D_B^T U^T B U D_B = \begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0_{n-r} \end{pmatrix}.$$

Let

$$P = UD_B.$$

Since both U and D_B are invertible, P is invertible.

Thus,

$$\boxed{P^T BP = J_{r,p},}$$

where

$$\boxed{J_{r,p} = \begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0_{n-r} \end{pmatrix}.}$$

Step 3. Apply the same construction to C.

Since C is also real symmetric, there exists an orthogonal matrix V such that

$$V^T CV = \operatorname{diag}(e_1, \ldots, e_n).$$

Because B and C have the same inertia, we may order the eigenvalues of C so that

$$\begin{cases} e_i > 0, & 1 \le i \le p, \\ e_i < 0, & p+1 \le i \le r, \\ e_i = 0, & r+1 \le i \le n. \end{cases}$$

Define

$$D_C = \operatorname{diag}\left(\frac{1}{\sqrt{|e_1|}}, \ldots, \frac{1}{\sqrt{|e_r|}}, 1, \ldots, 1 \right).$$

As before, D_C is invertible.

Setting

$$Q = VD_C,$$

we obtain

$$\boxed{Q^T CQ = J_{r,p}.}$$

Therefore, both matrices are congruent to the same diagonal matrix:

$$\boxed{P^T BP = Q^T CQ = J_{r,p}.}$$

Step 4. Construct the required congruence matrix.

From the preceding equality,

$$P^T BP = Q^T CQ.$$

Multiplying on the left by Q^{-T} and on the right by Q^{-1}, we obtain

$$Q^{-T} P^T BPQ^{-1} = C.$$

Rearranging the factors,

$$C = (PQ^{-1})^T B(PQ^{-1}).$$

Define

$$\boxed{S = PQ^{-1}}.$$

Since both P and Q are invertible, the matrix S is also invertible.

Consequently,

$$\boxed{C = S^T BS}.$$

Thus, B and C are congruent.

This completes the proof that equality of inertia implies congruence.

□

6.4 Conclusion of the Proof

We have now established both directions of Sylvester's Law of Inertia.

First, we proved that congruent real symmetric matrices necessarily have the same inertia.

Conversely, we demonstrated that two real symmetric matrices having the same inertia can each be transformed into the same canonical diagonal matrix, and are therefore congruent.

Hence,

$$\boxed{B \text{ and } C \text{ are congruent } \iff \text{In}(B) = \text{In}(C).}$$

7 Applications in Mathematical Statistics

Sylvester's Law of Inertia has important applications in mathematical statistics, where symmetric matrices arise naturally in the study of covariance, multivariate distributions, parameter estimation, and optimization.

In this section, we illustrate several connections between matrix congruence and statistical theory.

7.1 Covariance Matrices and Linear Transformations

Let X be an n-dimensional random vector with mean vector $\mu = \mathbb{E}[X]$ and covariance matrix

$$\Sigma_X = \mathbb{E}\left[(X - \mu)(X - \mu)^T\right],$$

assuming the required second moments exist.

Recall that every covariance matrix is symmetric and positive semidefinite. Indeed, for any vector $v \in \mathbb{R}^n$,

$$\begin{aligned} v^T \Sigma_X v &= \mathbb{E}\left[v^T (X-\mu)(X-\mu)^T v\right] \\ &= \mathbb{E}\left[\left(v^T (X-\mu)\right)^2\right] \\ &\geq 0. \end{aligned}$$

Consequently, all eigenvalues of Σ_X are nonnegative.

Now suppose we introduce a new random vector through an invertible linear transformation

$$Y = S^T X,$$

where $S \in \mathbb{R}^{n \times n}$ is invertible.

The mean of Y is

$$\mu_Y = S^T \mu.$$

Therefore, its covariance matrix is

$$\begin{aligned} \Sigma_Y &= \mathbb{E}\left[(Y-\mu_Y)(Y-\mu_Y)^T\right] \\ &= \mathbb{E}\left[S^T (X-\mu)(X-\mu)^T S\right] \\ &= S^T \mathbb{E}\left[(X-\mu)(X-\mu)^T\right] S. \end{aligned}$$

Hence,

$$\boxed{\Sigma_Y = S^T \Sigma_X S}.$$

This is precisely a congruence transformation.

By Sylvester's Law of Inertia,

$$\boxed{\mathrm{In}(\Sigma_Y) = \mathrm{In}(\Sigma_X).}$$

Since covariance matrices are positive semidefinite, they have no negative eigenvalues. Thus, if $\mathrm{rank}(\Sigma_X) = r$, then

$$\mathrm{In}(\Sigma_X) = (r, 0, n-r).$$

The transformed covariance matrix has the same inertia:

$$\mathrm{In}(\Sigma_Y) = (r, 0, n-r).$$

Therefore, an invertible linear transformation may change the individual variances, covariances, and eigenvalues, but it cannot change the rank of the covariance matrix or the number of its zero eigenvalues.

Example 7.1 (A Singular Covariance Matrix)**.** Consider the covariance matrix

$$\Sigma_X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its eigenvalues are 2 and 0, so

$$\mathrm{In}(\Sigma_X) = (1, 0, 1).$$

This covariance structure corresponds to two random variables having identical centered values almost surely.

Now choose the invertible matrix

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Under the transformation $Y = S^T X$, the covariance matrix becomes

$$\begin{aligned} \Sigma_Y &= S^T \Sigma_X S \\ &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}. \end{aligned}$$

The eigenvalues of Σ_Y are 5 and 0.

Although the nonzero eigenvalue has changed, the inertia remains

$$\boxed{\mathrm{In}(\Sigma_Y) = \mathrm{In}(\Sigma_X) = (1, 0, 1).}$$

Thus, the covariance matrices are congruent but have different eigenvalues, exactly as predicted by Sylvester's Law of Inertia.

Remark 7.2. In statistical applications, the rank of a covariance matrix represents the number of linearly independent directions having nonzero variance.

An invertible change of coordinates cannot create or eliminate such directions. It can only change how they are represented.

7.2 Whitening and Mahalanobis Distance

Another important application of congruence appears in the whitening of multivariate data.

Suppose that a random vector X has a positive definite covariance matrix Σ. By the spectral theorem,

$$\Sigma = Q \Lambda Q^T,$$

where all diagonal entries of Λ are strictly positive.

We may therefore define

$$\Sigma^{-1/2} = Q \Lambda^{-1/2} Q^T.$$

Consider the transformed random vector

$$Y = \Sigma^{-1/2}(X - \mu).$$

Its covariance matrix is

$$\begin{aligned}\mathrm{Cov}(Y) &= \Sigma^{-1/2}\Sigma\Sigma^{-1/2} \\ &= I.\end{aligned}$$

Hence,

$$\boxed{\mathrm{Cov}(Y) = I}.$$

This transformation is called *whitening*. It transforms the covariance matrix into the identity matrix.

For example, if

$$\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix},$$

then

$$S = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

satisfies

$$\boxed{S^T\Sigma S = I}.$$

Notice that the eigenvalues have changed from 4 and 9 to 1 and 1. Nevertheless, both matrices have inertia $(2, 0, 0)$.

Sylvester's Law of Inertia explains why this normalization is possible: every real symmetric positive definite matrix is congruent to the identity matrix.

An important consequence appears in the Mahalanobis distance, defined for a positive definite covariance matrix by

$$d_M^2(x, \mu) = (x - \mu)^T\Sigma^{-1}(x - \mu).$$

Under an invertible change of coordinates,

$$y = S^T(x - \mu), \qquad \Sigma_Y = S^T\Sigma S,$$

we obtain

$$\Sigma_Y^{-1} = S^{-1}\Sigma^{-1}S^{-T}.$$

Consequently,

$$\begin{aligned}y^T\Sigma_Y^{-1}y &= (x - \mu)^T S S^{-1}\Sigma^{-1}S^{-T}S^T(x - \mu) \\ &= (x - \mu)^T\Sigma^{-1}(x - \mu).\end{aligned}$$

Thus, the Mahalanobis distance is invariant under invertible linear changes of coordinates.

This illustrates a broader geometric principle: although a covariance matrix changes with the coordinate system, the statistical geometry it describes can remain unchanged.

7.3 Fisher Information and Reparameterization

A further application arises in statistical parameter estimation.

Let $\theta \in \mathbb{R}^n$ be a parameter vector and let $\ell(\theta; Z)$ denote the log-likelihood function associated with an observation Z.

The score vector is defined by

$$g_\theta = \nabla_\theta \ell(\theta; Z).$$

Under the usual regularity assumptions, the Fisher information matrix can be written as

$$I_\theta = \mathbb{E}[g_\theta g_\theta^T].$$

The Fisher information matrix is symmetric and positive semidefinite, since

$$v^T I_\theta v = \mathbb{E}[(v^T g_\theta)^2] \geq 0.$$

Now suppose that we introduce a new parameter vector ϕ through a smooth, invertible reparameterization

$$\theta = h(\phi).$$

Let

$$J = \frac{\partial \theta}{\partial \phi}$$

denote its Jacobian matrix, assumed invertible at the parameter value under consideration.

By the multivariable chain rule,

$$g_\phi = J^T g_\theta.$$

Consequently,

$$\begin{aligned} I_\phi &= \mathbb{E}[g_\phi g_\phi^T] \\ &= \mathbb{E}[J^T g_\theta g_\theta^T J] \\ &= J^T \mathbb{E}[g_\theta g_\theta^T] J. \end{aligned}$$

Therefore,

$$\boxed{I_\phi = J^T I_\theta J}.$$

Once again, we obtain a congruence transformation.

By Sylvester's Law of Inertia,

$$\boxed{\mathrm{In}(I_\phi) = \mathrm{In}(I_\theta).}$$

Thus, a smooth, locally invertible reparameterization preserves the rank and inertia of the Fisher information matrix.

In particular, if the Fisher information matrix is positive definite, it remains positive definite under such a reparameterization.

Likewise, if the Fisher information matrix is singular, an invertible reparameterization cannot remove that singularity.

This is important in parameter estimation because singular Fisher information can indicate a lack of local identifiability: certain parameter directions may not be distinguishable from the available statistical information.

7.4 Likelihood Curvature and Critical Points

The full significance of inertia becomes especially apparent when studying the curvature of statistical objective functions.

Suppose that $\ell(\theta)$ is a twice continuously differentiable log-likelihood function, and let

$$H_\theta = \nabla_\theta^2 \ell(\theta)$$

denote its Hessian matrix.

Since the Hessian is symmetric, its inertia describes the numbers of positive, negative, and zero curvature directions.

Now consider a smooth invertible reparameterization

$$\theta = h(\phi),$$

with Jacobian J.

At a stationary point, where

$$\nabla_\theta \ell(\theta) = 0,$$

the multivariable chain rule gives

$$\boxed{H_\phi = J^T H_\theta J.}$$

Thus, the Hessians in the two parameterizations are congruent at corresponding stationary points.

Sylvester's Law of Inertia therefore implies

$$\boxed{\mathrm{In}(H_\phi) = \mathrm{In}(H_\theta).}$$

This has an important interpretation.

If the Hessian is negative definite, the stationary point is a strict local maximum. If it is positive definite, the stationary point is a strict local minimum. If it has both positive and negative eigenvalues, the stationary point is a saddle point.

Because inertia is preserved under invertible reparameterization, these nondegenerate curvature classifications do not depend on the choice of parameter coordinates.

Remark 7.3. For nonlinear reparameterizations, the Hessian transformation generally contains additional terms involving first derivatives of the objective function. These terms vanish at stationary points, which is why the congruence relation above holds there.

The preceding examples illustrate how Sylvester's Law of Inertia provides a unifying principle across several areas of mathematical statistics.

Whether we are transforming covariance matrices, whitening multivariate data, reparameterizing statistical models, or examining the curvature of likelihood functions, congruence preserves fundamental structural information that is independent of the particular coordinate system.

8 Conclusion

In this note, we have explored the relationship between quadratic forms, matrix congruence, and Sylvester's Law of Inertia. Beginning with the distinction between similarity and congruence, we introduced the concept of inertia and established that the numbers of positive, negative, and zero eigenvalues of a real symmetric matrix remain invariant under congruence.

We then presented a detailed proof of Sylvester's Law of Inertia in both directions, using the spectral theorem, orthonormal eigenbases, and the rank-nullity theorem. Finally, we explored its applications in mathematical statistics, including covariance transformations, whitening, Fisher information, and likelihood optimization. The central lesson is that although individual eigenvalues may change under congruence, their distribution into positive, negative, and zero values remains unchanged.

It would be rather unfair to conclude this note without mentioning the remarkable mathematician whose name is associated with this theorem. James Joseph Sylvester (1814–1897) was an English mathematician who played a pioneering role in the development of modern matrix theory and helped establish matrix algebra as a mathematical subject in its own right. He introduced the term *matrix* in 1850, and his mathematical exchanges with Arthur Cayley contributed significantly to the emergence of modern linear algebra.

Interestingly, Sylvester's academic career was almost as extraordinary as his mathematics. Despite ranking second in the Cambridge Mathematical Tripos of 1837, he was initially denied a degree because his Jewish faith prevented him from subscribing to the religious requirements of the university.

His teaching career was hardly uneventful either. In 1841, Sylvester accepted a professorship at the University of Virginia, but his appointment ended rather dramatically in 1842 following a dispute involving student misconduct.

According to a famous anecdote, Sylvester once became involved in a confrontation with a student and struck him with a sword-cane. The student reportedly collapsed, leaving Sylvester briefly convinced that he had killed him!

More colorful versions of the story portray Sylvester as an exceptionally strict teacher, whose students were too frightened to ask questions. Indeed, according to such tales, the real danger was not asking a question, but being asked one by Sylvester himself. If the unfortunate student failed to answer, the punishment was supposedly rather more physical than a poor examination grade!

These stories even portray Sylvester as a mathematician repeatedly losing teaching positions because of his temperament and having to seek fresh appointments elsewhere. Although such embellishments are difficult to substantiate historically, his turbulent experience at Virginia and his

subsequent moves between England and America certainly make for an unusually eventful academic career.

In 1876, Sylvester joined the newly established Johns Hopkins University, where he helped develop advanced mathematical education and founded the American Journal of Mathematics. Among the younger scholars he mentored were Thomas Craig, Fabian Franklin, and Christine Ladd, later known as Christine Ladd-Franklin, who went on to distinguished intellectual careers. Particularly remarkable was his encouragement of Christine Ladd at a time when women were largely excluded from advanced mathematical education. His willingness to support and mentor a female mathematician was unusually progressive for his era and reveals another, perhaps less eccentric, side of Sylvester's personality.

One might therefore jokingly wonder whether Sylvester's legendary teaching methods actually worked. After all, several of his students became accomplished mathematicians. Perhaps a little fear of the professor was the secret ingredient—who knows!

Of course, the more convincing explanation is Sylvester's extraordinary mathematical insight, intellectual enthusiasm, and commitment to developing younger scholars.

Whatever the truth behind the more entertaining stories, Sylvester's lasting contribution lies in the mathematics he helped create. His Law of Inertia remains a beautiful example of how a seemingly simple observation about the signs of eigenvalues can reveal a profound and coordinate-independent mathematical structure.

Perhaps that is the most fitting conclusion: mathematicians, with all their eccentricities, eventually pass into history, but their mathematical ideas continue to educate generations long after they are gone.

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